EEE Previous Question Answers

AUT-23

1. a) $V1 = 12v, V2 = -6v$
   b) $i_0 = 6A, V_0 = 24v$
2. a) $V_0 = 20v$
   b) $i_1 = 2.085A, i_2 = 0.653A, i_3 = 1.231A$
3. a) $i = 3A,\ p = 72$
   or,
   a) $V_{th} = 9v, R_{th} = 3ohms\ ,\ I = 2.25A$
   b) $V_0 = 3.2v$
   or,
   b) $v_x = 25$

SP-23

1. b) $I = 4A,\ V_{ab} = 28v$
2. b) $V_1 = 80v, V_2 = -64v, V_3 = 156v$
   c) $i_1 = 2A,\ i_2 = 0A$
3. a) $I = .75A$
   b) $V_{th} = 9v, R_{th} = 3ohms, I = 2.25A$
   or
   a) $v_0 = 3.2v$
   b) $I_N = 1A, R_N = 4ohms$

AUT-22s

1. b) $19ohm$
   c) $R_{ab} = 40ohms,\ I = 2.5A$
2. a) $V_1 = 14v, V_2 = 22v$
   b) $I_1 = 25A, I_2 = 23A, I_3 = 20A$
   c) $I_1 = 0.8\ ohms, I_2 = 0.8\ ohms$

3. a) $I_3 = 4.5A$
   or,
   a) $I_1 = 1.67A, I_2 = \frac{1}{2}A$
   b) $V_1 = \frac{40}{3}v\ , V_3 = 20v$

SP-22

1. b) $I_1 = 2.73, I_2 = 0.27A, I_3 = 2.725 \times 10^{-3}A, I_4 = 2.723 \times 10^{-5}A$ [taking I = 3A, as my ID C231013]
   c) $R_{eq} = 10ohms$
2. a) $i_1 = -0.46A, i_2 = -0.52A, i_3 = -1.129A$ [replacing 16v to 13v]
   b) not done (super node)
3. $V_{th} = 10.66v, R_{th} = 888.8mOhms, I_N = ?\ , P_m = 32.03W$

~ [Prepared by Tanjim Hasan]